
UNIT 5 4G AND 5G NETWORKS

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5.0 INTRODUCTION

The mobile technologies such as MIMO, 4G, 5G have revolutionized the human experience. These mobile technologies have re-defined the way we communicate, collaborate and do business.

The advanced mobile technologies provide fast data access, low latency data transfer, high availability, better coverage and many other benefits. Starting with mobile radios we have come a far way wherein we now have a super-computer capability in our palms.

Modern mobile phones enable us to watch high definition live broadcast, multi-player games, video calling and many other applications.

In this chapter we discuss the technologies behind the 4G and 5G technologies along with its features, use cases and other examples.

5.1 OBJECTIVES

After going through this unit, you should be able to

- understand key concepts of High speed packet access,
- understand the usage of MIMO in High speed packet access,
- understand the key features of LTE and WiMAX 16E
- understand the key features of OFDM and LTE advanced,
- Understand the main features of 5G.

5.2 HIGH SPEED PACKET ACCESS

The high-speed packet access (HSPA) is the wireless technology to enhance the increase the upload and download speeds for 3G networks, UMTS and WCDMA networks. HSPA includes refers to High-Speed Downlink Packet Access (HSDPA) and High-Speed Uplink Packet Access (HSUPA). HSPA needs 3G network.

Modern application such as video streaming, live broadcast, real-time gaming application use HSPA.

Table 1 the key features of HSPA –

Table 1 HSPA Features

HSPA Feature	Brief Details
Peak Upload Speed	5.76MBPS
Peak Download Speed	14.4MBPS for 3G UMTS
Average Upload Speed	1-2 MBPS
Average Download Speed	3-5 MBPS
Round trip time	< 100ms

5.2.1 Working of HSPA

We have explained the working of HSPA below –

- The HSDPA and HSUPA provide low latency and high bit rates. Over 3G, the HSDPA creates a high-speed download channel among users.
- HSDPA uses 10ms transmission duration to enable quicker switching and accommodating the network changes. Faster data traffic scheduling is used to accelerate the data transfer for a single user.
- To efficiently use the network resources HSDPA uses QPSK/16 QAM modulation schemes are used. HSDPA uses various methods such as Short transmission time interval (TTI), Higher order modulation, Fast scheduling and user diversity and Fast link adaptation.
- HSDPA uses efficient retransmission of the missed data packets.
- HSUPA enhances the upload uses a dedicated channel between mobile phone and the station for the data upload.
- HSUPA also quickly retransmits the missed data packets.

The key features of HSDPA are Adaptive modulation and coding (AMC), Multi-code operation, Hybrid automatic repeat request, Higher order modulation, Short transmission time interval, Fast link adaptation and Fast scheduling.

5.2.2 High Speed Packet Access+

HSPA+ uses higher order modulation scheme (64-QAM for downlink and 16-QAM for uplink) and uses Multiple Input Multiple Output antenna technology and other enhancements.

Given below are the main features of HSPA+

HSPA Feature	Brief Details
Peak Download Data transfer	21MBPS with 64-QAM
Peak Download data rate	HSPA R9 downlink 126MBPS

Peak Upload data rate	HSPA R9 downlink 23MBPS
Modulation	64-QAM for downlink and 16-QAM for uplink

☛ Check Your Progress 1

1. The peak upload speed of HSPA is _____
2. The Average Download Speed of HSPA is _____
3. HSDPA uses _____ for efficient use of network resources.
4. The Peak Download Data transfer _____ in HSPA+ is _____ with 64-QAM
5. The Modulation used by HSPA+ is _____

5.3 MIMO IN HSPA

Multi Input Multi Output (MIMO) uses multiple antennas for transmitting and receiving to improve the data transfer rates.

The detailed working of MIMO in HSPA is given below –

- MIMO uses antennas and “processing” at both transmitter and receiver end to create multiple, uncorrelated radio streams between the transmitter and receiver.
- The radio streams use same frequency and time that multiplies the capacity in the same spectrum.
- As the antennas increase, the data transfer rates increase using the same amount of power and the spectrum.
- Each increase in the order of MIMO, we can double the data peak rate. For instance a 2x2 MIMO doubles the peak data rate over 1x1 MIMO.

We have depicted a 2x2 MIMO in Figure 1

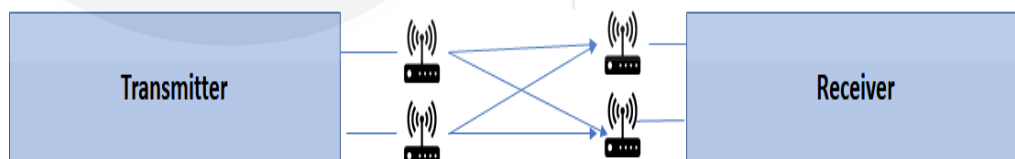


Figure 1 2x2 MIMO

As we have 2 antennas at transmitter and at the receiver end, the data transfer rate is doubled as compared to a 1x1 MIMO.

5.4 LTE AND WIMAX16E

LTE is the successor of UMTS and CMDA and provided 50 times better performance than existing cellular networks.

Long term evolution (LTE) was evolved as the mobile gaming, mobile video gained popularity. LTE provides high data rate, flexible bandwidth, packet optimizations and low latency. LTE also provides great quality of service. LTE uses both frequency

division duplex (FDD) and Time division duplex (TDD). FDD data uplink and data downlink uses different frequency and in TDD data uplink and data downlink are separated in time.

LTE also supports MIMO through which the base station transmits multiple data streams simultaneously.

Table 2 provides the key features of LTE

Table 2 LTE Features

Feature	Brief details
Peak Download Link	300MBPS
Peak Upload Link	75 MBPS
Frequency	1.4MHZ to 20 MHZ
Compatibility	GSM/EDGE/UMTS on 2G/3G spectrum.
Use cases	VoIP, live streaming, multimedia delivery, broadband internet

The architecture of LTE is depicted in Figure 2

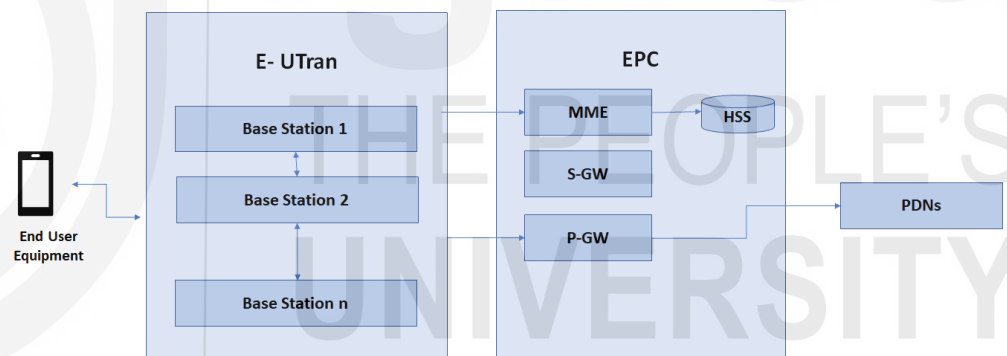


Figure 2 LTE Architecture

The end user equipment is a mobile device that handles the communication and data streams. The Universal Integrated Circuit Card (UICC) run Universal Subscriber Identity Module (USIM) application for managing the data such as phone number, security keys and others.

The E-UTRAN manages the communication between the mobile device and the EPC through base stations. Each base station provides the network for the mobile phone through one or more cells. The base station sends and receives the radio signals. The base station also sends the handover command for the mobile phones.

The evolved packet core network (EPC) manages the core functions and message routing. Given below are the key components in EPC –

- The mobility management entity (MME) manages the high level mobile operation by sending messages to Home Subscriber Server (HSS).
- The HSS database has the subscriber information.

- Packet Data Network (PDN) Gateway (P-GW) communicates with the packet data network (PDN).
- The serving gateway (S-GW) is the proxy for data transmission from the base station and P-GW.

5.4.1 WIMAX 16E

WIMAX is the broadband internet that supports various frequency bands. WIMAX uses sub-channels wherein the transmit channels are segregated into multiple parallel sub-channels that are assigned to the users. When the proximity of the user to the tower is high, the sub channels will be more.

The main features of WIMAX is given in table 3 -

Table 3 WIMAX Features

Feature	Description
Frequency band	Various frequency bands up to 11 GHZ
MIMO	2x2 MIMO
Modulation	QPSK for longer range modulation and 64 QAM for shorter range modulation
Security	AES based authentication
TDD	For uplink and downlink

Figure 3 WIMAX Features

5.4.2 Ultra-Wide Band

Ultra-wide band transmits the data with more than 500MHZ bandwidth spectrum. Ultra-wide band uses low energy for both high bandwidth and short-range communication.

Ultra-wide band uses short pulses to achieve greater location accuracy. An ultra-wide band can determine the location and distance between two ultra-wide band enabled devices with high degree of accuracy. For example, we can use Ultra-wide band for automatically opening the gate when the car approaches it. Another use case for ultra-wide band is that a smart phone can use short pulses to unlock the car door.

The main applications for ultra-wide band are mobile telephony, radar technology, wireless data transfer, real time location management, mobile TV, high speed Internet, video calls and digital security keys.

Table 4 provides the key features of Ultra-wide band

Table 4 Ultra-wide band features

Features	Description
Average data transfer rate	Downlink 5.9 MBPS
Peak Data downlink	326 MBPS for LTE 4 x 48
Peak Data uplink	88 MBPS for LTE 4 x 48
Bandwidth	500 MHZ
Latency	50 ms

5.4.3 Broadband Wireless Access

Broadband wireless provides a high speed internet access over a larger area. Wireless broadband uses radio waves to connect to the Internet backbone to provide high speed Internet access. Wireless broadband provides asymmetrical downlink and uplink data transfer rates.

Table 5 provides the main features of broadband wireless –

Table 5 Broadband wireless features

Features	Description
Average data transfer rate	~1.5MBPS
Data downlink	100 MBPS
Data uplink	50 MBPS
Bandwidth	More than 1 MHZ
Range	About 50KM

5.5 4G NETWORKS: HS-OFDM, LTE ADVANCED

4G networks support data transfer rate of more than 100MBPS. Due to this high data transfer rate, 4G is used for high definition video streaming, mobile gaming and such applications.

4G requires at least 40MHZ bandwidth for supporting the high data transfer rate. 4G also supports high mobility and roaming across various networks. The 4G system design requires LTE Advanced or WiMAX 402.16m

5.5.1 OFDM

Orthogonal Frequency Division Multiplex (OFDM) is a 4G signal format that provides multiple access schemes. OFDM uses multiple close spaced carriers each with low rate data stream. The transmitted data is shared across all carriers to provide high fault tolerance. In the multi-carrier scenario, OFDM uses code multiplexing and each of the carriers use mutually exclusive orthogonal codes.

For downlink it provides orthogonal frequency division multiple access (OFDMA) and for uplink it provides single channel orthogonal frequency division multiple access (SC-FDMA).

The usage of multiple carriers each with low data rate makes the OFDM resilient by shielding against narrow band fading.

Frequency Domain Equalization

By using appropriate frequency domain equalization at the receiver end, the direct sequence CDMA (DS-CDMA) provides high performance. Single carrier FDMA (SC-FDMA) technique uses frequency-division multiple access FDMA. The SC-FDMA encodes the sub-carriers.

MIMO

4G uses advanced multi-user MIMO in which multiple MIMO transceivers pre-code using signal processing algorithms.

5.5.2 LTE Advanced

LTE Advanced (LTE-A) is an enhancement over the LTE that provide higher data transfer rates, better performance mainly at the cell edges.

Table 6 provides the key features of LTE Advanced –

Table 6 LTE Advanced Features

Feature	Description
Peak data uplink	500MBPS
Peak data downlink	1 GBPS
Latency	< 5ms for one way packet transmission
User throughput	3x LTE
Use cases	Streaming media, High Definition Video streaming, Live TV

5.5.3 WIMAX 16M

WIMAX is the commercial name for Worldwide Interoperability for Microwave Access. WiMAX is a connection-oriented technology. The physical layer (PHY) and the media access control (MAC) layer are standardized by the WifiMax technology.

The physical layer operates in the bandwidth between 1.25MHZ and 20 MHZ and uses OFDMA for data transfer. The physical layer also supports MIMO antennas for high performance.

The MAC describes the technologies such as Internet protocol (IP), ethernet, asynchronous transfer mode (ATM) along with methods for data security, data transfer, data classification, secure communication, encryption methods (such as AES, DES encryption standards).

Table 7 provides the main features of WiFiMax –

Table 7 WiFIMax features

Feature	Description
Bandwidth	1GBPS
MIMO	Single user MIMO and multi-user MIMO
Multiplexing	TDD and FDD
Frequency	450-470 MHZ to 3.600 GHZ
Use cases	Broadband internet, High Definition Video streaming, Mobile Internet, Multimedia streaming.

Check Your Progress 2

1. The Increase in the order of MIMO, _____ the data peak rate.
2. The peak download link of LTE is ____
3. The frequency of LTE is _____.
4. _____ in LTE manages the core functions and message routing.
5. The Data downlink of broadband wireless is _____
6. The Frequency of WiFiMax is _____

5.6 FEATURES OF 5G NETWORKS

5G is the next generation of wireless technology that aims to provide high data transfer rate. 5G leverages many technologies such as MIMO, Software defined network, Internet of Things (IoT), thousands of simultaneous hyperlink connections, cloud computing and others. 5G enables enterprises to securely share the data and services among its employees and customers.

5G uses shorter millimeter waves between 30GHz and 300GHz enabling faster data transfer rates and provides ultra-low latency, high reliability, increase network capacity and rich user experiences. 5G delivers 1000 times more capacity than 4G.

5G also supports many use cases such as real time video streaming, big data processing, digital media services, connected vehicles, autonomous driving, virtual reality, telesurgery, multi-player gaming, immersive shopping, faster cloud access and others.

Table provides the key 5G features –

Feature	Description
Latency	1 ms
Data transfer rate	10 GBPS
Availability	99.999%
Network energy consumption	90% reduction in network energy consumption

5.7 SUMMARY

The high-speed packet access (HSPA) is the wireless technology to enhance the increase the upload and download speeds for 3G networks, UMTS and WCDMA networks. The HSDPA and HSUPA provide low latency and high bit rates. Over 3G, the HSDPA creates a high-speed download channel among users. The key features of HSDPA are Adaptive modulation and coding (AMC), Multi-code operation, Hybrid automatic repeat request, Higher order modulation, Short transmission time interval, Fast link adaptation and Fast scheduling. Multi Input Multi Output (MIMO) uses multiple antennas for transmitting and receiving to improve the data transfer rates. LTE is the successor of UMTS and CMDA and provided 50 times better performance than existing cellular networks. WiMAX is the broadband internet that supports various frequency bands. Ultra-wide band transmits the data with more than 500MHz bandwidth spectrum. Ultra-wide band uses low energy for both high bandwidth and short-range communication. 4G networks support data transfer rate of more than 100MBPS. Orthogonal Frequency Division Multiplex (OFDM) is a 4G signal format that provides multiple access schemes. WiMAX is a connection-oriented technology with the physical layer (PHY) and the media access control (MAC) layer are standardized by the WifiMax technology. 5G uses shorter millimeter waves between 30GHz and 300GHz enabling faster data transfer rates and provides ultra-low latency.

5.8 SOLUTIONS/ANSWERS

Check Your Progress 1

1. 5.76MBPS
2. 3-5 MBPS

3. QPSK/16 QAM modulation schemes
4. 21MBPS
5. 64-QAM for downlink and 16-QAM for uplink

Check Your Progress 2

1. doubles
2. 300 MBPS
3. 1.4MHZ to 20 MHZ
4. evolved packet core network (EPC)
5. 450-470 MHZ to 3.600 GHZ

5.9 FURTHER READINGS

References

Mobile Computing 3rd Edition by Raj Kamal - <https://www.amazon.in/Mobile-Computing-Raj-Kamal/dp/0199455414>